A Comprehensive Survey on Attention Mechanisms in Modern Deep Learning
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Section 1. Introduction to Attention and Sequence Modeling
Traditional Recurrent Neural Networks (RNNs) and LSTMs suffered from catastrophic forgetting and bottlenecke
hidden states when processing long sequences. The introduction of the Transformer architecture by Vaswani et
revolutionized sequence modeling by replacing recurrence entirely with Scaled Dot-Product Attention.
Attention computes a weighted representation where queries (Q), keys (K), and values (V) interact via matrix mu

Section 2. Multi-Head Attention Architecture
Rather than computing a single attention function, Multi-Head Attention projects queries, keys, and values
into h distinct subspaces. This allows the model to jointly attend to information from different representation
subspaces at different positions. Mathematical formulation: Attention(Q, K, V) = softmax((Q K^T) / sqrt(d_k)) V.

Attention Mechanisms Survey: Optimization & Benchmarks

Section 3. Optimization Techniques and Learning Dynamics
Training large-scale transformer architectures requires specialized optimization regimes:
1. AdamW Optimizer: Decoupled weight decay regularization to stabilize gradient updates.
2. Warmup Schedules: Linear warmup for the initial 2,000 steps followed by cosine annealing.
3. Precision: Mixed precision training (bfloat16 and FP8) significantly decreases memory footprints and increases

Section 4. Empirical Evaluation Benchmarks
Standardized language evaluation relies on MMLU (Massive Multitask Language Understanding),
GSM8K for grade-school mathematical reasoning, and HumanEval for code generation capabilities.
Recent scaling laws indicate compute optimal frontier follows Chinchilla scaling (Hoffmann et al., 2022).